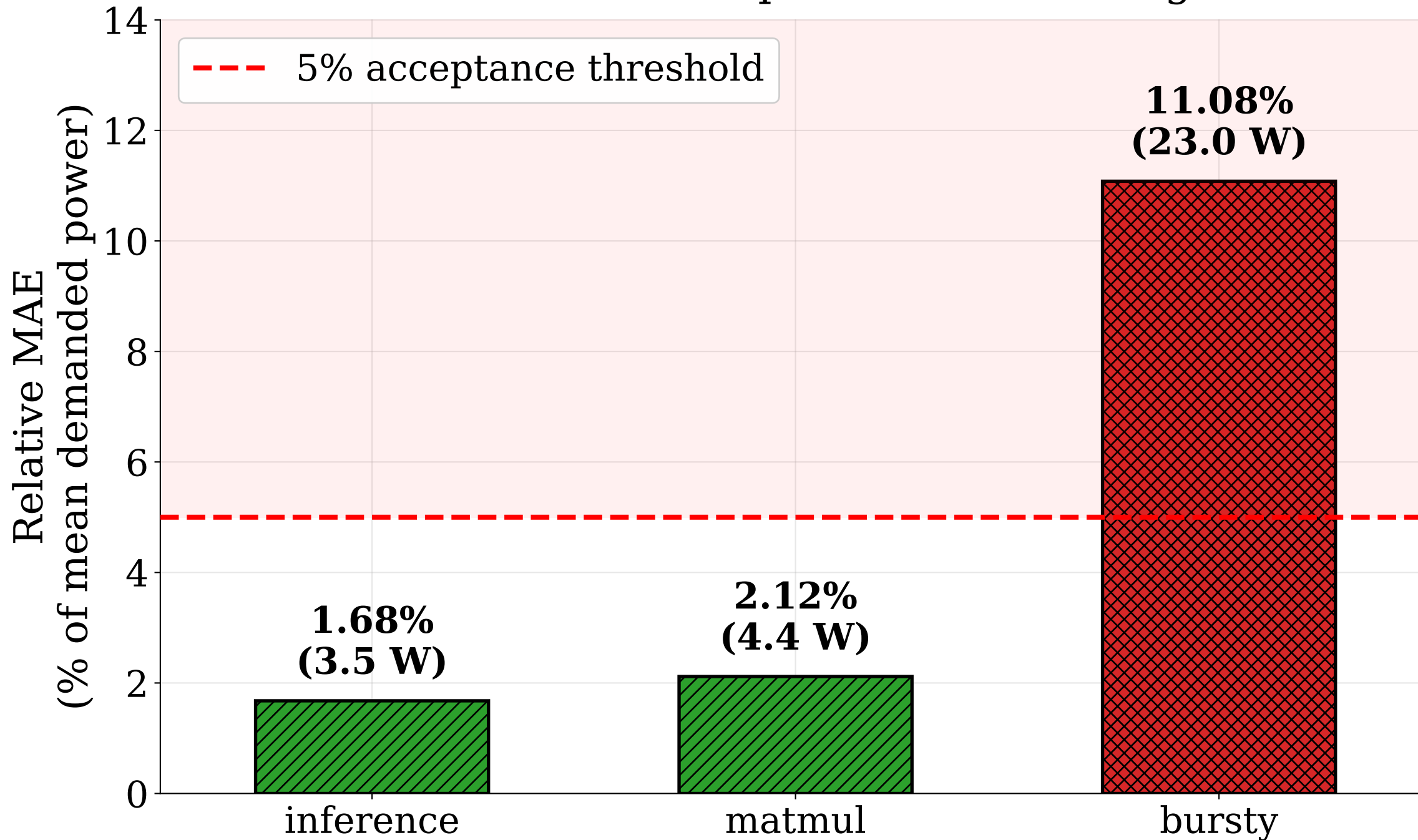


E4 — Closed-loop demand-following



Below 5% threshold

inference: 1.68% | matmul: 2.12%

Tier-1 inner-loop PID at 200 Hz tracks the demand profile within the cell-PUE design margin. The relative MAE is well within the classical control-engineering tolerance band.

Above 5% threshold

bursty: 11.08%

Workload bimodality (compute/idle period-T cycles) exceeds the inner-loop PID time constant. The multi-tier cascade absorbs the residual at Tier-2 (host AR(4), 1 Hz). This is the binding constraint that motivates the cascade design.